

```
$ ps -ef
```

Process Utilities

Managing, monitoring, and controlling running processes from the Linux shell: ps, w, jobs, kill, nice, and renice.

What Is a Process?

A process is **a program in execution** — once a program is loaded into memory, the kernel gives that running copy a unique identity (PID) and its own dedicated resources (memory, CPU time, open files), completely independent of every other process, including other copies of the same program.

PROGRAM

A static file on disk (e.g. /usr/bin/chrome) — just code and data, doing nothing until run.



PID

Unique number the kernel assigns to every running process.



UID

Identifies which user owns and can control the process.

PROCESS

That program loaded into memory and running — has a PID, its own memory, and CPU time.



PPID

The PID of the process that created (forked) this one.



Priority / Nice

How much CPU time the scheduler gives the process.

Example: open Chrome and it becomes PID 4521. Open a second Chrome window — that's a new process, PID 4522, with its own memory, even though it's the same program.

Process Lifecycle

Created (fork)



Ready



Running



Waiting / Sleeping



Terminated

The ps Command

“Process Status” — snapshot of currently running processes

`ps`

Shows processes running in the current shell/session only

`ps -e` (or `ps -A`)

Shows every process running on the system

`ps -f`

Full-format listing — includes PPID, start time, command

`ps -ef`

Full listing of every process on the system (most used)

`ps aux`

BSD-style listing — includes %CPU and %MEM usage

`ps -u <user>`

Shows processes owned by a specific user

Note: `ps -ef` and `ps aux` both list every process, just in different formats — `ps -ef` is UNIX-style (PPID, STIME), `ps aux` is BSD-style (%CPU, %MEM).

Reading ps -ef Output

UID	PID	PPID	C	STIME	TTY	TIME	CMD
root	842	1	0	09:12	?	00:00:01	/usr/sbin/sshd

UID User who owns / started the process	PID Unique process identification number	PPID Parent process's PID	C CPU utilization factor
STIME Time the process started	TTY Terminal associated with the process	TIME Cumulative CPU time used	CMD Command / program that launched it

Note: a CMD ending in <defunct> marks a zombie process — it has finished executing but its parent hasn't yet read its exit status.

Who's on the System: w



Shows who is logged in and what each user is currently doing.

Displays:

- Username and terminal (TTY)
- Login time and idle time
- Remote host (if logged in remotely)
- Current command each user is running
- System uptime and load average (in header)

```
$ w
```

Note: finger is largely deprecated / disabled by default on modern distros for security reasons.

Foreground & Background Jobs

Controlling how processes run in the shell

&

Run a command in the background

```
sort bigfile.txt &
```

jobs

List background jobs in the current shell

```
jobs -l
```

fg

Bring a background job to the foreground

```
fg %1
```

bg

Resume a stopped job in the background

```
bg %1
```

Ctrl + Z

Suspend (pause) the running foreground job

—

Ctrl + C

Terminate the running foreground job

—

Note: %1 refers to job number 1 from jobs; a backgrounded job that tries to read input auto-suspends until brought to the foreground with fg.

Ending & Prioritizing Processes

`kill` · `nice` · `renice`

`kill`

Sends a signal to a process, most often to terminate it.

`kill PID`

Sends SIGTERM — asks the process to terminate gracefully

`kill -9 PID`

Sends SIGKILL — forces immediate termination

`kill -l`

Lists all available signal names

`nice` / `renice`

Control how much CPU priority a process gets. Range: -20 (highest) to 19 (lowest).

`nice -n 10 cmd`

Starts a new command with a lower (nicer) priority

`renice -n 5 -p PID`

Changes the priority of an already running process

*Only root can set a negative (higher) priority.
Regular users may only lower a process's priority.*

Note: `kill` doesn't delete a process from disk — it only signals the kernel to stop the running instance in memory.

Recap: Process Utilities



`ps`

View a snapshot of running processes



`w`

See who is logged in and what they're doing



`finger`

Get detailed info about a specific user



`& / jobs / fg / bg`

Move jobs between foreground and background



`kill`

Send a signal to stop or control a process



`nice / renice`

Set or change a process's CPU priority

Next up: Disk Utilities — `du`, `df`, `mount`, `umount`, `find`, `unlink`, `umask`, `ulimit`